Superstore Sales Analysis

Insights and Recommendations

This document presents seven business insights found in the Superstore sales data (9,800 order lines, 4,922 orders, 793 customers, January 2015 to December 2018, total sales $2.26 million). Each insight is backed by the SQL analysis and followed by practical recommendations. All findings are about **revenue**, because the dataset has no profit, discount or quantity columns.

Summary

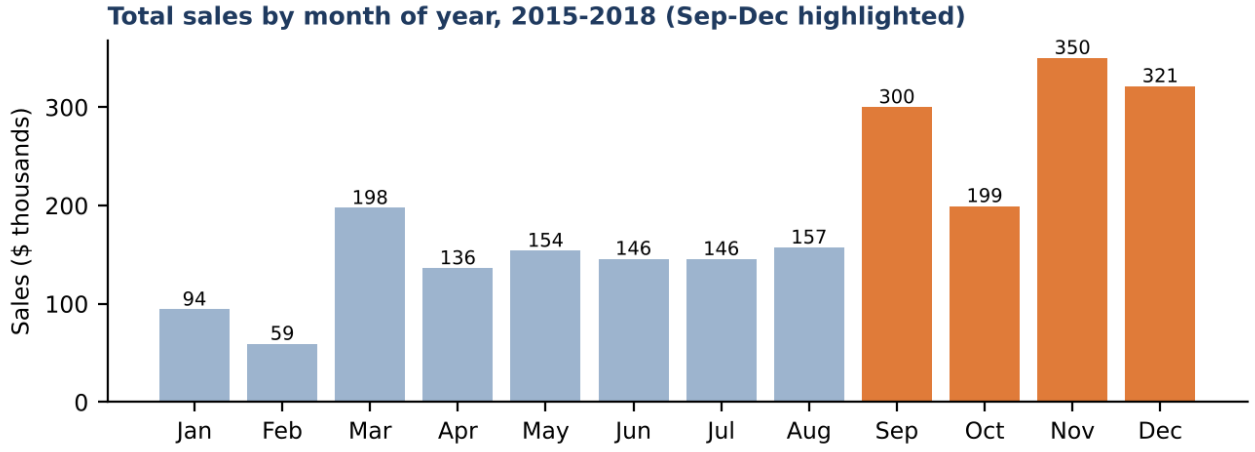
#	Insight	Recommendation	Priority
1	September to December bring in 51.8% of yearly sales; January and February only 6.8%.	Build stock and campaigns for Sep-Dec; run offers in Jan-Feb.	High
2	Orders grew 75% from 2015 to 2018, but average order value fell 14%.	Raise basket size with bundles, cross-selling and a free-shipping threshold.	High
3	The top 8 sub-categories make 78.4% of sales; 4.7% of lines (those above $1,000) make 43.2%.	Protect stock of top items; review the 4 smallest sub-categories using profit data.	Medium
4	California and New York make 33.2% of sales; 26 of 49 states are under 1% each; Central fell in 2018.	Protect CA and NY; run targeted campaigns in Central, South and small states.	High
5	The top 20% of customers generate 48.5% of sales; only 13 of 793 customers bought once.	Create a VIP or loyalty programme and personal service for the top customers.	High
6	Home Office sales grew 51% in 2018; Corporate customers spend the most per customer.	Create business offers: volume pricing and bundles for Corporate and Home Office.	Medium
7	Standard Class is used for 60% of orders; every ship mode delivers on its promise.	Keep standard delivery reliable and promote faster options for urgent orders.	Low

Priority is a suggestion based on how much revenue is involved and how quickly the business could act.

Insight 1: Sales are highly seasonal: September to December is the main season

What the data shows

Over the four years, September to December produced 51.8% of all sales, and November and December alone produced 29.7%. November is the strongest single month (15.5%). January and February together produce only 6.8% of sales, with February the weakest month at 2.6%. The same pattern repeats every year, so it is a real seasonal cycle and not a one-off.



Source: SQL queries Q3 and Q4.

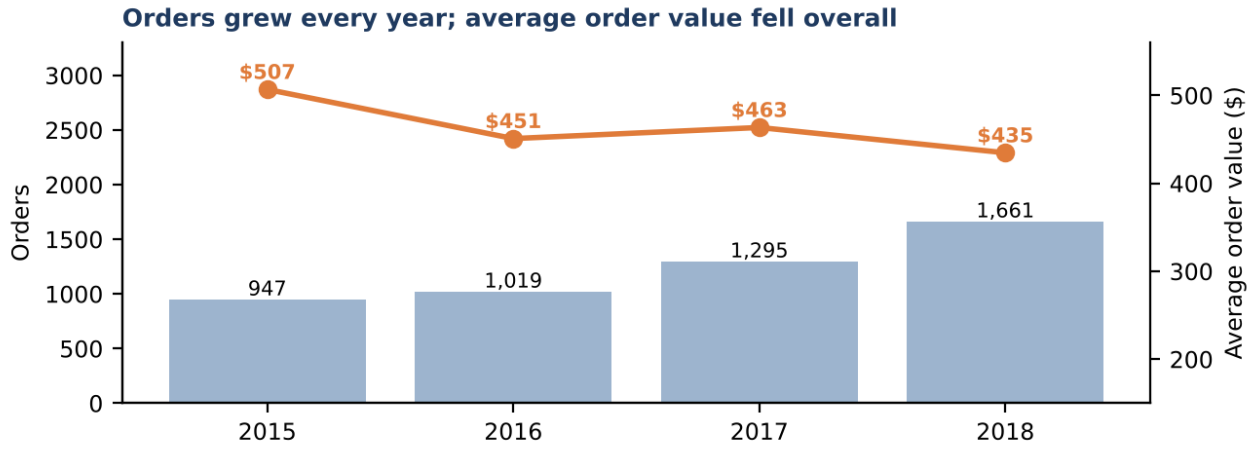
Recommendations

- Order stock and plan delivery and staffing capacity by August so the company can handle the Sep-Dec peak without shortages or delays.
- Time the biggest marketing campaigns and promotions for September to November, when customers are already buying.
- Run targeted offers in January and February (for example discounts for business customers or bundle deals) to lift the weakest months.

Insight 2: Growth comes from more orders, but each order is getting smaller

What the data shows

Sales rose from $479,856 in 2015 to $722,052 in 2018, with growth of 30.6% in 2017 and 20.3% in 2018. This growth came from volume: orders increased from 947 to 1,661 (+75%), and active customers from 589 to 690. At the same time the average order value fell from $507 in 2015 to $435 in 2018 (-14%). The business is winning more orders but earning less from each one.



Source: SQL queries Q2 and Q1, plus yearly order and customer counts from the same table.

Recommendations

- Offer product bundles that pair Office Supplies (high volume, low value) with Technology or Furniture (high value).
- Introduce a free-shipping or discount threshold slightly above the current average order value, to encourage larger baskets.
- Track average order value as a monthly KPI next to total sales.

Insight 3: A small number of sub-categories and big-ticket items drive most revenue

What the data shows

The top 8 of 17 sub-categories (led by Phones, Chairs and Storage) bring in 78.4% of sales, while the four smallest (Art, Envelopes, Labels, Fasteners) add up to only 2.6%. The 462 order lines worth $1,000 or more are just 4.7% of all lines but produce 43.2% of sales. A single product, the Canon imageCLASS 2200 copier, sold $61,600 from only 5 lines (2.7% of total sales). Copiers as a whole grew almost 6 times, from $10,850 in 2015 to $62,899 in 2018. In contrast, Machines fell 22% in 2018 (from $55,907 to $43,545) while most other main sub-categories grew.

Source: SQL queries Q5 to Q8, plus sub-category sales by year.

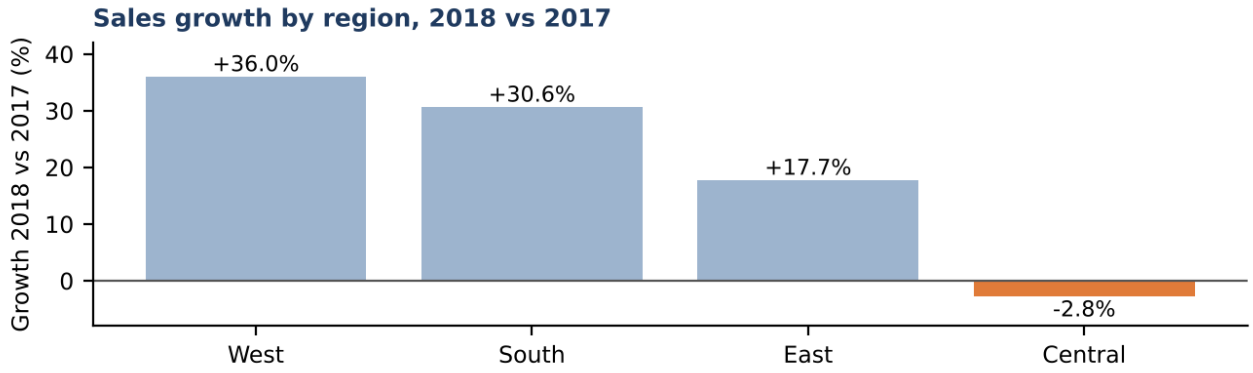
Recommendations

- Keep the best-selling and high-value products (phones, chairs, storage, copiers) well stocked, and make sure large orders are processed and delivered reliably.
- Invest in the fast-growing Copiers sub-category and investigate why Machines declined in 2018 (pricing, stock-outs or competition).
- Before removing the four small sub-categories, check their profit and their role in attracting customers. This dataset has no profit column, so it cannot decide this alone.

Insight 4: Sales depend on a few states, and Central and South are lagging

What the data shows

California (19.7%) and New York (13.5%) together account for 33.2% of all sales. At the other end, 26 of the 49 states each contribute less than 1% of sales (for example North Dakota, West Virginia and Maine). By region, West is largest (31.4%) and South smallest (17.2%). In 2018, West grew 36.0%, South 30.6% and East 17.7%, but Central fell 2.8%. South also had a sharp drop of 32% in 2016, which shows it is the least stable region.



Source: SQL queries Q9 to Q11, plus regional sales by year.

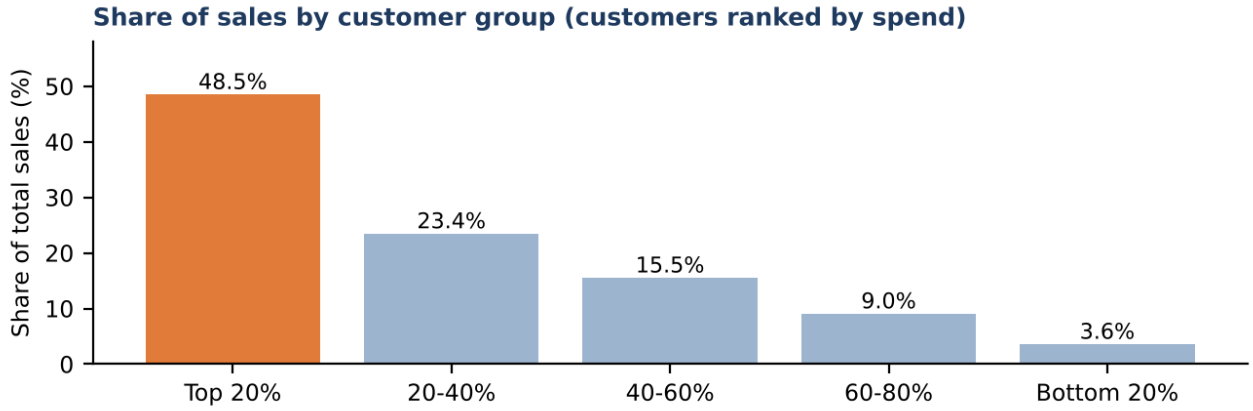
Recommendations

- Protect the two core states with reliable delivery and good service, because losing a large share of either would hurt the whole business.
- Find out why Central declined in 2018 and run a recovery plan there (local promotions, review of top accounts).
- Pilot low-cost marketing in a few of the smallest states to test whether there is real demand.

Insight 5: A top group of customers drives half of revenue, and the base is loyal

What the data shows

The top 20% of customers (159 people) generate 48.5% of sales, while the bottom 20% generate only 3.6%. The ten best customers each spent more than $12,000, led by Sean Miller with $25,043. The customer base is also very loyal: only 13 of 793 customers placed a single order, and 449 placed more than five orders. This means the business keeps customers well, and the main opportunity is to get more from the valuable ones and grow the lower-spending ones.



Source: SQL queries Q13 to Q15.

Recommendations

- Launch a loyalty or VIP programme for the top 20%, with early access to offers and a named account contact for the top 10.
- Monitor top customers closely, for example flag any who have not ordered for several months, because losing even a few would have a large effect.
- Encourage the middle groups to spend more with personalised product suggestions based on what they already buy.

Insight 6: Corporate and Home Office customers are growing fastest

What the data shows

Consumers are the biggest segment with 50.8% of sales (409 customers). However, the business segments are growing faster: in 2018 Home Office sales grew 51.2% and Corporate 15.2%, compared with 12.9% for Consumers. Corporate customers also spend the most per customer ($2,917, against $2,872 for Home Office and $2,807 for Consumers), yet there are only 236 of them.

Source: SQL query Q12, plus segment sales by year.

Recommendations

- Offer volume discounts, regular-order plans and bundles (furniture plus technology plus supplies) for Corporate and Home Office customers.
- Add outreach to small businesses and home-office workers, since this segment is growing fastest.
- Keep Consumers engaged with the seasonal promotions, because they are still half of revenue.

Insight 7: Shipping works as promised, and Standard Class is the default choice

What the data shows

Ship modes perform as expected: Same Day orders ship in about 0 days, First Class in 2.2 days, Second Class in 3.3 days and Standard Class in 5.0 days. The longest time to ship in the data is 7 days (Standard Class). Most customers choose the standard option: 59.8% of orders (2,945 of 4,922) use Standard Class. Reliable delivery is a strength of the business.

Source: SQL query Q16.

Recommendations

- Keep Standard Class dependable, especially in the Sep-Dec peak when volume is highest.
- Promote First Class and Same Day for urgent or high-value orders, such as technology purchases.
- Use the fast and reliable delivery as a selling point in marketing.

Limitations and next steps

- **No profit data.** Revenue is not profit. A sub-category, region or customer with high sales may still be unprofitable, so confirm important decisions with profit and discount data before acting.
- **Results are from the data as loaded.** The cleaning step (Task 2) may change a few rows, such as 11 missing postal codes and one near-duplicate line. Re-run the SQL queries after cleaning to confirm the figures.
- **Four years of history.** Trends such as the 2017 and 2018 growth should be monitored, not assumed to continue.

KPIs to track going forward: monthly sales and September to December share, average order value, sales growth by region (especially Central and South), revenue from the top 20% of customers, and Standard Class delivery time.